MILITARY SPECIFICATION

PLUG, TEST, ATOMIC WEAPON

1. SCOPE

1.1 This specification covers special requirements, quality assurance provisions, and the preparation for delivery criteria for one type of Test Plug used on the Atomic Demolition Charge.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-S-60902(MU) - Soldering, Electrical Connections for Nuclear Weapons Systems, Procedure for

STANDARDS

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices
MIL-STD-1169(MU) - Packaging, Packing and Marking for Shipments of Inert Ammunition Components

DRAWINGS

ORDNANCE CORPS

8808362 - Packing and Marking for Storage and Shipments
8861386 - Plug, Test, Atomic Weapon
8861390 - Connector, Plug, Electrical
8864250 - Potting Boot; Marking
8883535 - Bag, Packing for Repair Parts
9207508 - Primer, Polyurethane Molding Compound
ET-8861386 - Equipment Tabulation

(Copies of specifications, standards, drawings, and publication required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 1195

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3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The test plug shall comply with all requirements specified on Drawing (dwg.) 8861386, with all requirements specified herein, and with the requirements of applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Functioning (see dwg 8861386).

3.4.1 Forward voltage drop. The forward voltage drop of each diode shall not exceed 1.0 volt with a constant current of 400 ± 5 milliamperes (ma) when tested as specified in 4.4.1.1.

3.4.2 Reverse leakage. The reverse leakage current of each diode shall not exceed 2 microamperes with a reverse voltage of 300 ± 5 volts direct current (vdc) when tested as specified in 4.4.1.2.

3.5 Insulation resistance. The insulation resistance of any circuit shall not be less than 20 megohms when subjected to a potential of 500 ± 50 vdc with the current limited to 5 ma when tested as specified in 4.4.2.

3.6 Environmental testing. The item shall comply with 3.4 and 3.5 before and after each environmental test when tested as specified in 4.4.3. Tests shall be conducted in the order specified.

3.6.1 Low temperature. The item shall withstand exposure to a storage temperature of -65°F and shall comply with 3.4 and 3.5 while at -65°F when tested as specified in 4.4.3.1.

3.6.2 High temperature. The item shall withstand exposure to a temperature of +160°F and shall comply with 3.4 and 3.5 while at +160°F when tested as specified in 4.4.3.2.

3.6.3 Vibration. The item shall withstand vibration when tested as specified in 4.4.3.3.

3.6.4 Humidity. The item shall withstand exposure to a humid atmosphere when tested as specified in 4.4.3.4.

3.7 Workmanship.

3.7.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristics.

3.7.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.7.3 Burrs. Parts shall be free from burrs.

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3.7.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.7.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

3.7.6 Electrical. Ends of wire leads shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted (see 6.4). When the contract requires first article inspection, the contractor shall submit 1 complete assembly (see dwg. 8861386), 1 subassembly before potting (see dwg. 8861386) and 1 set of parts, for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in table I. Tests need not be conducted in order specified in table I.

Table I
First article inspection

Examination or test	Contractor's inspection permissible before submission	Destructive	Non-Dest	Assembly	Sub-assembly	Part
1. Classification of characteristics (see 4.3.2.1)	X		X	X	X	X
2. Functioning (see 3.4)	X		X	X		
3. Insulation resistance (see 3.5)	X		X	X		
4. Vibration (see 3.6.1)		X		X		
5. High temperature (see 3.6.2)	X		X	X		
6. Low temperature (see 3.6.3)	X		X	X		
7. Humidity (see 3.6.4)		X		X		

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4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Product acceptance inspection.

4.3.1 Lot formation. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies, or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table and inspected for all characteristics in the applicable classification. Any characteristic in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	309	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	2362- or more	298
138-139	123	253-255	176	531-540	229		

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Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics. Critical, major, and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Connector, plug, electrical (see dwg 8861390); covering details of dwg 8861386.

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Objective evidence (see 6.2) available for component in accordance with applicable drawing, specification or standard	Visual
Minor:		
201.	Workmanship (see 3.7)	Visual-Manual

4.3.2.1.2 Plug, test, atomic weapon (see dwg 8861386) before potting.

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major.		
101.	Objective evidence (see 6.2) for diodes in accordance with dwg 8861386	Visual
102.	Height of diodes	Gage
103.	Soldering in accordance with MIL-S-60902(MU)	Visual-Manual
104.	Objective evidence (see 6.2) for thread in accordance with dwg 8861386	Visual
Minor:		
201.	Workmanship (see 3.7)	Visual-Manual

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4.3.2.1.3 Plug, test, atomic, weapon (see dwg 8861386).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Potting in accordance with dwg 8861386	Visual
102.	Length from top of potting to top of boot before adding thread	Gage
103.	Length of thread spiraled in potting	*Measure
104.	Length of thread after potting	*Measure
105.	Knot coated with compound 8881768	Visual
106.	Eye of thread loop free of compound	Visual
* Measure with suitable scale		
Minor:		
201.	Marking in accordance with dwg 8864250	Visual

4.3.2.1.4 Production parts preparation for delivery (see 5.1).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Packaging, packing and marking in accordance with MIL-STD-1169(MU)	Visual
Minor:		
201.	Workmanship (see 3.7)	Visual

4.3.2.1.5 Bag, Packing, for Repair Parts (see dwg 8883535 covering dwg. part number 8883535-7 and section 5.2).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Material (see 3.1)	Visual
102.	Length before packing, min	*Measure
103.	Width before packing, min	*Measure
104.	Objective evidence (see 6.2) available for heat seal test in accordance with MIL-P-116	Visual

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105. Properly packed.....Visual
106. No tear, split, or crack in bag.....Visual

* Measure with suitable scale.

Minor:
201. Marking.....Visual
202. Workmanship (see 3.7).....Visual-Manual

4.3.2.1.6 Repair part packing and marking for storage and shipments (see dwg. 8808362 and section 5.2).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Gross load weight, max.....*Measure
102. Contents tight in package.....Manual
103. Correctly packed.....Visual/Manual
104. Package material (see 3.1).....Visual

* Determine weight with a suitable scale.

Minor:
201. Marking.....Visual
202. Workmanship (see 3.7).....Visual-Manual

4.3.3 Testing. All the tests specified in this paragraph shall be performed.

	Test	Test Classification
a.	Functioning	Major
b.	Insulation resistance	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Functioning (see 3.4).

4.3.3.1.1 Forward voltage drop (see 3.4.1). Each diode in each assembly shall be tested as specified in 4.4.1.1 If any diode fails to comply with 3.4.1, the assembly shall be rejected and removed from the lot.

4.3.3.1.2 Reverse leakage (see 3.4.2). Each diode of each assembly shall be tested as specified in 4.4.1.2. If any diode fails to comply with 3.4.2, the assembly shall be rejected and removed from the lot.

4.3.3.2 Insulation resistance (see 3.5). The insulation resistance of each test plug shall be tested as specified in 4.4.2. If any test plug fails to comply with 3.5, the test plug shall be rejected and removed from the lot.

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4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). EL's for assemblies are listed on the Equipment Tabulation, ET-8861386. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment. The contractor shall obtain approval of such designs, operating instructions, and special calibration instructions as specified in 6.3 prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired.
b. The code "CD" is listed in the "Part of Identifying Number" column on the EL.

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on EL's. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Functioning (see 3.4).

4.4.1.1 Forward voltage drop (see 3.4.1). The forward voltage drop shall be tested with approved test equipment in accordance with table IV.

Table IV
Forward voltage drop

Positive Connector Pin	Negative Connector Pin
B	A
E	C

4.4.1.2 Reverse leakage (see 3.4.2). The reverse leakage shall be tested in accordance with MIL-STD-750, method 4016.2 and table V.

Table V
Reverse leakage test

Postive Connector Pin	Negative Connector Pin
C	E
A	B

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4.4.2 Insulation resistance (see 3.5). The test plug insulation resistance shall be tested with approved insulation measuring equipment in accordance with table VI.

Table VI
Insulation resistance (see dwg 8861386)

Postive potential applied to connector pin	Minus potential to all other pins and shell except pins
A	B, C, and E

4.4.3 Environmental testing (see 3.6).

4.4.3.1 Low temperature (see 3.6.1). The item shall be tested in accordance with MIL-STD-810, Method 502.

4.4.3.2 High temperature (see 3.6.2). The item shall be tested in accordance with MIL-STD-810, Method 501, Procedure I.

4.4.3.3 Vibration (see 3.6.3). The item shall be tested in accordance with MIL-STD-810, Method 514, Procedure X. Test curve shall be AB, Fig. 514-6. The item shall not be operated during the test.

4.4.3.4 Humidity (see 3.6.4). The item shall be tested in accordance with MIL-STD-810, Method 507, Procedure I.

5. PREPARATION FOR DELIVERY

5.1 Production items.

5.1.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1.1 Level A. The test plug shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.1.2 Level C. The test plug shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Packing.

5.1.2.1 Levels A and C. The test plug shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.1.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

5.2 Repair parts.

5.2.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.2.1.1 Level A. The test plug shall be packaged in accordance with dwgs 8883535 and 8808362 using the specified moisture barrier.

5.2.1.2 Level C. The test plug shall be packaged in accordance with dwg 8808362 except that the moisture barrier may be omitted.

5.2.2 Packing.

5.2.2.1 Level A and C. The test plug shall be packed in accordance with dwgs 8883535 and 8808362.

5.2.3 Marking. Marking shall be as specified in accordance with dwgs 8883535 and 8808362, as applicable.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Production or repair items.
c. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment. The contractor shall obtain approval of equipment designs, operating instructions, and calibration instructions from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-NR4.

6.4 Instructions for first article inspection (see 4.2.1). Parts may be used in subassemblies if the examination or tests (see 4.2.2, Table I) are verified by a representative designated by the Government. Subassemblies may be used in final assemblies if the examination or tests (see 4.2.2, Table I) are verified by a representative designated by the Government.

Custodian:
Army - MU

Preparing activity:
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Project No. 1195-A059

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-P-50293(MU) Plug,Test, Atomic Weapon

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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